Math 105L **Lecture 4-2**

Rates of Change: From Average Velocity to Instantaneous Velocity

1. The Geometry of Average Velocity

The formula for average velocity over a time interval $[a, b]$ has the exact mathematical form of a line slope:

$$\text{Average Velocity on } [a,b] = \frac{\Delta s}{\Delta t} = \frac{s(b) - s(a)}{b - a}$$

Setting $b = a + h$, the interval is $[a, a + h]$ and the average velocity becomes the **difference quotient**:

$$\text{Avg Vel on } [\mathbf{a}, \mathbf{a+h}] = \frac{\mathbf{s(a+h) - s(a)}}{\mathbf{h}}$$

Geometric Interpretation: The average velocity equals the slope of the **secant line** connecting $(a, s(a))$ and $(a + h, s(a + h))$.

2. Classroom Case Study: Projectile Motion $s(t) = -4t(t-1)$

Consider an object moving along a line with position $s(t) = -4t(t-1) = -4t^2 + 4t$ (in meters) from $t = 0$ to $t = 1$ seconds:

- **Interval** $[0, 1]$**:**

$$\text{Avg Vel} = \frac{s(1) - s(0)}{1 - 0} = \frac{0 - 0}{1} = \mathbf{0} \text{ m/s} \quad \text{(secant line is horizontal)}$$

- **Interval** $[0, 0.5]$**:**

$$\text{Avg Vel} = \frac{s(0.5) - s(0)}{0.5 - 0} = \frac{1 - 0}{0.5} = \mathbf{2} \text{ m/s}$$

- **Interval** $[0, 0.1]$**:**

$$s(0.1) = -4(0.1)(-0.9) = 0.36 \implies \text{Avg Vel} = \frac{0.36 - 0}{0.1} = \mathbf{3.6} \text{ m/s}$$

- **Interval** $[0, 0.01]$**:**

$$s(0.01) = -4(0.01)(-0.99) = 0.0396 \implies \text{Avg Vel} = \frac{0.0396 - 0}{0.01} = \mathbf{3.96} \text{ m/s}$$

The Instantaneous Velocity at $t = 0$ As the time interval h shrinks toward zero, the secant lines pivot toward the **tangent line** at $(0, 0)$:

$$v(0) = \lim_{h \to 0} \frac{s(0+h) - s(0)}{h} = \lim_{h \to 0} \frac{-4h(h-1)}{h} = \lim_{h \to 0} \big[-4(h-1) \big] = \mathbf{4} \text{ m/s}$$

3. Formal Summary

- **Instantaneous Velocity Definition:**

$$\mathbf{v(a)} = \lim_{\mathbf{h \to 0}} \frac{\mathbf{s(a+h) - s(a)}}{\mathbf{h}}$$

- **Graphical Signature:** On the graph of position $s(t)$, the instantaneous velocity at $t = a$ is given by the **slope of the tangent line** to the curve at the point $(a, s(a))$.